Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Bengaluru, India

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Summary

Software Engineer with a strong algorithmic foundation – LeetCode Guardian (2166 peak, top 1.1%, 1300+ problems, 107 contests), Google Kick Start Round E '22 Rank 1289, 500+ GfG problems, HackerRank 6-Star Problem Solving. Two years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, shipping DevOps platforms for enterprise clients (RWS, State Street, DTCC) and an MLOps pipeline AWS published as a sample. Deep across Python, TypeScript, React, FastAPI, AWS (SageMaker, ECS, Lambda), Terraform, and AI-lab-adjacent tooling (Claude Code plugins, MCP servers on Bedrock, Feast, Airflow). 11 upstream PRs merged (axios, Airflow, lucide, Prefect, Feast, awslabs/mcp, multer, and others). 5x AWS certified.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Bengaluru, India

- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Cut per-account setup from 3+ days to under 4 hours (~90%) by replacing a legacy account-vending pipeline with Terraform for a data-centre exit and a second landing zone: 4,000+ resources across 20+ Terraform Cloud workspaces, none failing.
  - Delivered two Transit Gateway hubs in separate AWS Regions with Network Firewall inspection, plus self-service onboarding that gives each new workload its route table within 15 minutes.
  - Won approval for AWS IPAM by measuring real billing in a sandbox (about 94% lower projected cost). Used Kiro and Claude Code for specs and design documentation.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization, with organization-wide auto-enable in Terraform. Project closed with a 10/10 CSAT.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC with Amazon Q.
- **MLOps Pipeline (SME Program, Jun – Aug 2025):** Built end-to-end SageMaker medical image classification pipeline with blue-green endpoint deployment, auto-scaling, and drift detection – zero-downtime model updates with automatic rollback. Provisioned via Terraform + CodePipeline; CloudFront frontend + API Gateway + Lambda inference. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Internal Impact:** Co-authored an AWS Prescriptive Guidance pattern on secure PR previews and approval-gated releases for Amplify Gen 2.

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy for zero-downtime releases. Established distributed tracing with X-Ray and CloudWatch.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD; deployed SonarQube on EC2 + RDS to embed continuous code-quality analysis into the delivery pipeline.

Publications & Talks

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **Org-wide tech talks (AWS internal):** Terraform testing with tftest + PyTest (2025); GitOps configuration safety with health checks and pipeline gating (2026). Associate Speaker Certified (2026).
- **Programs (AWS internal):** MLOps SME Program Graduate (2025). 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless).

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Projects

itr-agent Jul 2026
TypeScript, MCP Protocol, Node.js, Zod GitHub
◇ Local-first MCP server for Indian income tax: the LLM interviews and calls 12 tools, while a deterministic engine over a versioned JSON rule pack computes every rupee. 151 tests pin it to published worked examples and the statute, on an Ubuntu and Windows CI matrix.

Kinfolk Jul 2026
TypeScript, Next.js 16, React 19, Drizzle, Neon Postgres GitHub | Live
◇ Collaborative family graph modelled as a directed acyclic graph, not a tree: each relative keeps their own record and consent links join them, so nothing is overwritten. Live on Vercel + Neon with a production health check; regression tests run on an in-memory PGlite database.

Claude Cost Optimizer Mar 2026
Claude Code Plugin, MCP, TypeScript, React, Python GitHub | Live
◇ Claude Code plugin, MCP cost server, CLI grader and web calculator for Claude spend. Rebuilt every estimator on cache-aware billing, correcting a modelled 30-turn Opus session from $3.44 to $0.74, and replaced every vague savings range with a sourced or computed number. Golden-value tests pin the cost math in all three implementations.

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

Other Notable Projects: Ledger Sync (self-hosted finance app, 26 pages), GitScope (Chrome Web Store extension), Blue-Green AWS Terraform (15 modules, ECS canary), Instagram Likes Leaderboard (browser-console tool), SelfHub (MCP personal-memory server).

Open Source Contributions (11 merged upstream)

- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, HCL, Bash
Systems & Backend: FastAPI, Node.js, Express, gRPC, REST, GraphQL, PostgreSQL, MongoDB, Redis, SQLite, Kafka
ML/AI: PyTorch, TensorFlow, Scikit-learn, SageMaker, Bedrock, MCP (Model Context Protocol), Claude Agent SDK, Feast, LangChain
Cloud & Infra: AWS (ECS, Lambda, SageMaker, API Gateway, CloudFront, IAM, Organizations, Transit Gateway), Terraform, Docker, GitHub Actions, CodePipeline
Frontend: React, Preact, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)
CS Fundamentals: Data Structures, Algorithms, System Design, Distributed Systems, OS, DBMS, Computer Networks

Competitive Programming

- **LeetCode Guardian** – 2166 peak rating (top 1.1%), 1300+ problems, 107 contests *Sagargupta1610*
- **Google Kick Start Round E '22** – Rank 1289 *Certificate*
- **GeeksforGeeks** 500+ DSA problems; **HackerRank** 6-Star Problem Solving, 5-Star C++